

INSURANCE CHARGES PREDICTION

1. The client requirement is very clear - to predict Insurance Charges. The dataset has inputs as numbers and output is also provided. So, this is Machine Learning, Supervised learning with Regression method.
2. The dataset has 5 inputs and 1 output column - 'charges', and 1338 rows.
3. There are 3 numeric columns and 2 categorical columns in the inputs given. These categorical columns - 'sex (Male / Female)' and 'smoker (Yes / No)' are nominal columns. So, we need to convert these strings to numbers using One Hot Encoding.

Algorithms Used

1. Multi Linear Regression

R2 Value - 0.7153639826310382

2. Decision Tree Regression

S No	Criterion	Max Features	Splitter	R2 Value
1	squared_error	None	best	0.6554698147987423
2	squared_error	None	random	0.6896713921766076
3	squared_error	Log2	best	0.7162107505398796
4	squared_error	Log2	random	0.628192984691905
5	squared_error	sqrt	best	0.7162107505398796
6	squared_error	sqrt	random	0.628192984691905
7	friedman_mse	None	best	0.6527621081821988
8	friedman_mse	None	random	0.6870679557260886
9	friedman_mse	Log2	best	0.7162107505398796
10	friedman_mse	Log2	random	0.6170997564430872
11	friedman_mse	sqrt	best	0.7162107505398796
12	friedman_mse	sqrt	random	0.6170997564430872
13	absolute_error	None	best	0.6951462144767093
14	absolute_error	None	random	0.6692181738482165
15	absolute_error	Log2	best	0.7112570609533369
16	absolute_error	Log2	random	0.6405230890154472
17	absolute_error	sqrt	best	0.7112570609533369
18	absolute_error	sqrt	random	0.6405230890154472
19	poisson	None	best	0.6679422851659789
20	poisson	None	random	0.6881390045964151
21	poisson	Log2	best	0.7093419711519469
22	poisson	Log2	random	0.6281984709873736
23	poisson	sqrt	best	0.7093419711519469
24	poisson	sqrt	random	0.6281984709873736

The Decision Tree Regression

R2 value (mse & friedman_mse, log2 & sqrt, best) - 0.7162107505398796

3. Random Forest Regression

S No	Criterion	Max Features	N_estimators	R2 Value
1	squared_error	None	100	0.8224670087103022
2	squared_error	None	500	0.825709750395295
3	squared_error	None	1000	0.8252230899169606
4	squared_error	Log2	100	0.836636928920273
5	squared_error	Log2	500	0.8385116520701519
6	squared_error	Log2	1000	0.8380745014951104
7	squared_error	sqrt	100	0.836636928920273
8	squared_error	sqrt	500	0.8385116520701519
9	squared_error	sqrt	1000	0.8380745014951104
10	friedman_mse	None	100	0.8226842196285769
11	friedman_mse	None	500	0.8258539946516261
12	friedman_mse	None	1000	0.8253582883045352
13	friedman_mse	Log2	100	0.8370040333442393
14	friedman_mse	Log2	500	0.8386565231589508
15	friedman_mse	Log2	1000	0.8381090803579065
16	friedman_mse	sqrt	100	0.8370040333442393
17	friedman_mse	sqrt	500	0.8386565231589508
18	friedman_mse	sqrt	1000	0.8381090803579065
19	absolute_error	None	100	0.8293605458335819
20	absolute_error	None	500	0.8296268814230406
21	absolute_error	None	1000	0.8291872017428092
22	absolute_error	Log2	100	0.8390658109974172
23	absolute_error	Log2	500	0.8408036400896118
24	absolute_error	Log2	1000	0.841080337979374
25	absolute_error	sqrt	100	0.8390658109974172
26	absolute_error	sqrt	500	0.8408036400896118
27	absolute_error	sqrt	1000	0.841080337979374
28	poisson	None	100	0.8256708221884399
29	poisson	None	500	0.826598376230095
30	poisson	None	1000	0.8267536496162815
31	poisson	Log2	100	0.8375503218155933
32	poisson	Log2	500	0.8379046466619692
33	poisson	Log2	1000	0.8383142176607175
34	poisson	sqrt	100	0.8375503218155933
35	poisson	sqrt	500	0.8379046466619692
36	poisson	sqrt	1000	0.8383142176607175

The Random Forest Regression

R2 Value (absolute_error, log2 & sqrt, 1000) - 0.841080337979374